Accuracy of Rise Velocity Estimation from Lagrangian Particle Tracking Data: A Comparative Analysis of Scattered and Interpolated Approaches

Abstract

In Lagrangian particle tracking (LPT) simulations of multiphase flows, field quantities such as gas volume fraction and velocity are carried by discrete particles that move to irregular, scattered positions over time. A fundamental question is how accurately one can recover integral flow diagnostics—in particular the bubble rise velocity—from these scattered Lagrangian data, compared with the reference solution on the original Eulerian grid. This document presents a systematic numerical analysis of rise velocity estimation for a two-dimensional axisymmetric rising bubble at Re = 100, Bo = 100. Two definitions of rise velocity (centroid-based and volume-averaged) are evaluated under four interpolation scheme combinations (linear/cubic for velocity and gas fraction) and two data representations (raw scattered particles versus re-interpolation onto a fixed Eulerian grid). The analysis reveals that: (i) the centroid-based rise velocity is consistently more accurate than the volume-averaged approach; (ii) re-interpolating scattered data onto a fixed grid dramatically improves centroid-based estimates while offering limited benefit for the volume-averaged method; and (iii) the dominant source of error is the velocity field interpolation rather than the gas fraction field.

1 Introduction

Consider a gas bubble rising through a quiescent liquid, a canonical problem in multiphase fluid dynamics. When solving the governing equations on a fixed Eulerian grid, the gas volume fraction field $\alpha_{\mathrm{g}}(\boldsymbol{x}, t)$ and the velocity field $\boldsymbol{V}(\boldsymbol{x}, t)$ are naturally available at every grid node. Computing integral diagnostics such as the bubble rise velocity from these gridded fields is straightforward.

An alternative Lagrangian viewpoint tracks a cloud of neutrally buoyant marker particles that are initially seeded at every Eulerian grid node, giving a total of $n_r \times n_z$ particles (where n_r and n_z denote the number of grid points in the radial and axial directions, respectively). Each particle carries the local gas fraction α_{g} and axial velocity V_z that are interpolated from the Eulerian field to the particle position at every time step. Because the particles are advected by the flow, their positions become irregularly distributed (scattered) over time.

While this Lagrangian representation is advantageous for data-driven modal decomposition methods such as Proper Orthogonal Decomposition (POD) and Dynamic Mode Decomposition (DMD), it introduces interpolation errors and raises the question: *how accurately can the bubble rise velocity be recovered from scattered Lagrangian particle data?*

Three principal sources of error are identified:

(i) **Interpolation scheme:** During particle tracking, the Eulerian fields are interpolated to each particle position using either linear or cubic interpolation, for both the velocity and gas fraction fields independently.

(ii) **Data representation:** The rise velocity can be computed either directly from the scattered particle data or after re-interpolating the scattered data back onto a fixed Eulerian grid.

(iii) **Rise velocity definition:** Two physically motivated definitions are available—centroid-based and volume-averaged—each with different sensitivities to field errors.

This document provides a detailed mathematical description of each approach (section 2), describes the numerical setup (section 3), and analyses the results (section 4).

2 Mathematical Formulation

2.1 Problem setting

We consider a two-dimensional axisymmetric domain parameterised by the radial coordinate r and the axial coordinate z. The Eulerian grid has n_r points in the r-direction and n_z points in the z-direction, forming a structured mesh $\{(r_i, z_j)\}$. At each time step t^n, the Eulerian solution provides:

- $\alpha_{\mathrm{g}}(r_i, z_j, t^n)$: the gas volume fraction, and
- $V_z(r_i, z_j, t^n)$: the axial (vertical) velocity component.

A set of $N_p = n_r \times n_z$ neutrally buoyant Lagrangian particles is initialised at the Eulerian grid nodes. At subsequent times, each particle p has a displaced position $\boldsymbol{x}_p(t) = (r_p, z_p)$ and carries scalar values $\alpha_{\mathrm{g}}^{(p)}$ and $V_z^{(p)}$ obtained by interpolating the Eulerian fields to the particle location.

2.2 Interpolation during particle tracking

At each time step of the tracking algorithm, the Eulerian fields $\alpha_{\mathrm{g}}(\boldsymbol{x}, t)$ and $V_z(\boldsymbol{x}, t)$ must be evaluated at the particle position $\boldsymbol{x}_p$. Two interpolation schemes are employed:

Linear interpolation. The field value at $\boldsymbol{x}_p$ is obtained by bilinear interpolation from the four surrounding Eulerian grid nodes. For a scalar field ϕ:

$$\phi(\boldsymbol{x}_p) = \sum_{i=1}^{4} w_i \, \phi(\boldsymbol{x}_i), \qquad \sum_{i=1}^{4} w_i = 1, \tag{1}$$

where w_i are the standard bilinear weights based on the normalised distances within the bounding cell.

Cubic interpolation. The field is evaluated using a bicubic interpolation stencil involving the 4×4 surrounding grid nodes, providing higher-order accuracy at the cost of a wider stencil.

Since the velocity and gas fraction fields can use different interpolation schemes independently, four combinations arise:

Case	α_{g} interpolation	V_z interpolation
G-linear / V-linear	Linear	Linear
G-linear / V-cubic	Linear	Cubic
G-cubic / V-linear	Cubic	Linear
G-cubic / V-cubic	Cubic	Cubic

2.3 Rise velocity definitions

Two definitions of the bubble rise velocity are considered. In an axisymmetric domain, all integrals below are area-weighted by the radial coordinate r.

2.3.1 Method 1: Centroid-based rise velocity

The centroid-based rise velocity is obtained by first computing the α_{g}-weighted centroid of the bubble in the axial direction, and then differentiating its position in time.

Step 1: Bubble centroid. On the structured Eulerian grid, the axial position of the bubble centroid at time t^n is defined as

$$z_c(t^n) = \frac{\displaystyle\sum_{i=1}^{n_r}\sum_{j=1}^{n_z} \alpha_{\mathrm{g}}(r_i, z_j, t^n)\, z_j}{\displaystyle\sum_{i=1}^{n_r}\sum_{j=1}^{n_z} \alpha_{\mathrm{g}}(r_i, z_j, t^n)} \,. \tag{2}$$

For the scattered Lagrangian data (without re-gridding), the equivalent expression uses the particle-carried quantities directly:

$$z_c(t^n) = \frac{\displaystyle\sum_{p=1}^{N_p} \alpha_{\mathrm{g}}^{(p)}(t^n)\, z_p(t^n)}{\displaystyle\sum_{p=1}^{N_p} \alpha_{\mathrm{g}}^{(p)}(t^n)} \,. \tag{3}$$

Step 2: Rise velocity via finite differencing. The centroid-based rise velocity is computed as

$$V_{\mathrm{rise}}^{\mathrm{cent}}(t^n) = \frac{z_c(t^n) - z_c(t^{n-1})}{t^n - t^{n-1}} = \frac{z_c(t^n) - z_c(t^{n-1})}{\Delta t} \,, \tag{4}$$

where Δt is the time step between snapshots. This method relies *solely* on the gas volume fraction field; the velocity field is not used. Consequently, any error in α_{g} interpolation directly affects the centroid position and, through finite differencing, the estimated rise velocity. This is implemented in the function `rise_velocity_centroid`, which returns both $z_c(t^n)$ and $V_{\mathrm{rise}}^{\mathrm{cent}}(t^n)$.

2.3.2 Method 2: Volume-averaged rise velocity

The volume-averaged rise velocity computes the gas-fraction-weighted, radially-weighted mean of the axial velocity inside the bubble region.

Step 1: Bubble identification. The bubble region Ω_b is defined by a threshold on the gas volume fraction:

$$\Omega_b(t^n) = \left\{ \boldsymbol{x} : \alpha_{\mathrm{g}}(\boldsymbol{x}, t^n) \geq \alpha_{\mathrm{th}} \right\}, \qquad \alpha_{\mathrm{th}} = 0.5 \,. \tag{5}$$

Step 2: Weighted velocity average. On the structured Eulerian grid, the volume-averaged rise velocity is

$$V_{\mathrm{rise}}^{\mathrm{vol}}(t^n) = \frac{\displaystyle\sum_{\substack{(i,j)\\ \in \Omega_b}} \alpha_{\mathrm{g}}(r_i, z_j, t^n)\; r_i\; V_z(r_i, z_j, t^n)}{\displaystyle\sum_{\substack{(i,j)\\ \in \Omega_b}} \alpha_{\mathrm{g}}(r_i, z_j, t^n)\; r_i} \,, \tag{6}$$

where the weighting by r_i accounts for the axisymmetric geometry (each grid cell at radial position r represents an annular volume proportional to r). Grid points outside the bubble ($\alpha_\mathrm{g} < \alpha_\mathrm{th}$) are excluded from the summation by zeroing their radial weight.

For scattered Lagrangian data, the same formula applies with particles replacing grid nodes:

$$V_\mathrm{rise}^\mathrm{vol}(t^n) = \frac{\displaystyle\sum_{\substack{p \\ \alpha_\mathrm{g}^{(p)} \geq \alpha_\mathrm{th}}} \alpha_\mathrm{g}^{(p)}\, r_p\, V_z^{(p)}}{\displaystyle\sum_{\substack{p \\ \alpha_\mathrm{g}^{(p)} \geq \alpha_\mathrm{th}}} \alpha_\mathrm{g}^{(p)}\, r_p}\,. \tag{7}$$

This method uses *both* the gas fraction field (for bubble identification and weighting) *and* the velocity field (the quantity being averaged). It is therefore sensitive to errors in both fields. This is implemented in the function `rise_velocity_mean`.

Comparison of the two methods. Table 1 summarises the key differences.

Table 1: Comparison of rise velocity estimation methods.

	Centroid-based	**Volume-averaged**
Fields required	α_g only	α_g and V_z
Error sources	α_g interpolation	α_g and V_z interpolation
Temporal derivative	Yes (finite diff.)	No
Spatial averaging	Implicit (centroid)	Explicit (weighted mean)

2.4 Data representation: Scattered vs. interpolated

After the Lagrangian tracking step, the particle positions are scattered throughout the domain. Two strategies are considered for post-processing:

Approach A: Direct use of scattered data. The rise velocity is computed directly from the particle-carried values $\{\alpha_\mathrm{g}^{(p)}, V_z^{(p)}, r_p, z_p\}$ using eqs. (3) and (7). No additional interpolation is performed.

Approach B: Re-interpolation onto a fixed Eulerian grid. The scattered particle data are first interpolated back onto the original Eulerian grid using a natural-neighbour scattered interpolant. Specifically, a `scatteredInterpolant` (with natural-neighbour method and 'none' extrapolation) is constructed for both α_g and V_z:

$$\hat{\alpha_\mathrm{g}}(r_i, z_j) = \mathcal{I}\big[\,\{r_p, z_p, \alpha_\mathrm{g}^{(p)}\}_{p=1}^{N_p}\,\big](r_i, z_j)\,, \qquad \hat{V_z}(r_i, z_j) = \mathcal{I}\big[\,\{r_p, z_p, V_z^{(p)}\}_{p=1}^{N_p}\,\big](r_i, z_j)\,, \tag{8}$$

where $\mathcal{I}[\cdot]$ denotes the natural-neighbour interpolation operator. Points outside the convex hull of the particle cloud receive a value of zero. The rise velocity is then computed from the re-gridded fields $\hat{\alpha_\mathrm{g}}$ and $\hat{V_z}$ on the structured grid using eqs. (2) and (6).

This re-gridding step introduces an additional interpolation error but restores the regularity of the data, potentially improving the accuracy of integral diagnostics.

2.5 Error metric

The accuracy of each Lagrangian rise velocity estimate is quantified by the root-mean-square (RMS) error relative to the Eulerian reference, normalised by the gravitational velocity scale $V_s = \sqrt{g\,D}$:

$$E_{\mathrm{RMS}} = \sqrt{\frac{1}{N_t}\sum_{n=1}^{N_t}\left(\frac{V_{\mathrm{rise}}^{\mathrm{Lag}}(t^n) - V_{\mathrm{rise}}^{\mathrm{Eul}}(t^n)}{V_s}\right)^2}, \tag{9}$$

where N_t is the number of time steps considered and $V_{\mathrm{rise}}^{\mathrm{Eul}}$ denotes the rise velocity computed from the original Eulerian DNS data.

3 Numerical Setup

The test case is a single gas bubble rising in a quiescent liquid under the combined action of buoyancy and surface tension, characterised by:

- Reynolds number: $\mathrm{Re} = 100$,
- Bond number: $\mathrm{Bo} = 100$,
- Bubble diameter: $D = 0.63$ m.

The simulation is performed on a 2D axisymmetric Eulerian grid. A total of $N_t = 450$ snapshots are recorded at a time step of $\Delta t = 0.01307$ s. Quantities are non-dimensionalised using the gravitational velocity scale $V_s = \sqrt{gD}$ and the corresponding time scale $\tau_s = D/V_s$, giving a dimensionless time horizon of $\tau \approx 25$.

The Lagrangian particles are initialised at every Eulerian grid node, yielding $N_p = n_r \times n_z$ particles. The tracking is performed with four combinations of interpolation schemes as described in section 2.2, producing four distinct sets of scattered particle trajectories and field values in each case.

4 Results and Discussion

4.1 Analysis I: Scattered Lagrangian data

In this first analysis, the rise velocity is computed directly from the raw scattered particle data (Approach A in section 2.4), without any re-interpolation onto a fixed grid.

4.1.1 Rise velocity time histories

Figure 1 shows the temporal evolution of the rise velocity for the first interpolation case (G-linear, V-cubic), comparing the Eulerian reference (red) with the Lagrangian scattered estimate (blue), for both the centroid-based (solid lines) and volume-averaged (dashed lines) methods.

The centroid-based Lagrangian rise velocity (blue solid) follows the Eulerian reference (red solid) closely throughout the simulation. In contrast, the volume-averaged Lagrangian estimate (blue dashed) exhibits significant and persistent deviations from its Eulerian counterpart (red dashed), particularly during the initial transient ($\tau < 5$) and in the quasi-steady regime ($\tau > 10$).

4.1.2 RMS error across interpolation combinations

Figure 1 also presents the RMS error (eq. (9)) for all four interpolation combinations, arranged in a heatmap. Two key observations emerge:

(1) **Centroid-based is more accurate:** The RMS errors for the centroid-based method ($E_{\mathrm{RMS}} \approx 8.7 \times 10^{-3}$) are consistently about 3.4 times smaller than those for the volume-averaged method ($E_{\mathrm{RMS}} \approx 2.9 \times 10^{-2}$), across all four interpolation combinations.

(2) **Weak dependence on interpolation scheme:** The choice of linear versus cubic interpolation for either α_{g} or V_z has a negligible effect on the RMS error when using scattered data directly. All four centroid-based errors fall within the narrow range $[8.70, 8.79] \times 10^{-3}$, and all volume-averaged errors lie in $[2.93, 2.98] \times 10^{-2}$.

The fact that the volume-averaged rise velocity is less accurate is consistent with its dependence on both α_{g} and V_z: errors in either field degrade the estimate. The centroid-based method, which depends only on α_{g}, is more robust.

4.1.3 Field profile diagnostics

To identify whether the gas fraction or the velocity field carries the dominant error, fig. 2 shows profiles of α_{g} and V_z along horizontal (r-direction) and vertical (z-direction) cut-lines at a representative time instant ($\tau \approx 6.3$, snapshot index 120).

The gas volume fraction profiles (fig. 2, middle panel) show that the scattered Lagrangian α_{g} values from all four interpolation combinations closely follow the Eulerian reference, with only minor scatter near the bubble interface where the gas fraction transitions sharply from $\alpha_{\mathrm{g}} \approx 1$ to $\alpha_{\mathrm{g}} \approx 0$.

In contrast, the velocity profiles (fig. 2, right panel) reveal substantially larger deviations. The scattered Lagrangian velocity values show significant spread around the Eulerian reference, particularly in the radial direction and near the bubble boundary. This confirms that *the velocity field interpolation is the dominant source of error* in the Lagrangian framework, which explains why the volume-averaged method (which directly uses V_z) is less accurate than the centroid-based method (which uses only α_{g}).

4.2 Analysis II: Re-interpolated Lagrangian data

In this second analysis, the scattered Lagrangian data are re-interpolated onto the original Eulerian grid (Approach B in section 2.4) before computing the rise velocity.

4.2.1 Rise velocity time histories

Figure 3 shows the rise velocity comparison after re-interpolation. Two striking improvements are observed compared with the scattered-data analysis:

(1) The centroid-based Lagrangian rise velocity (blue solid) now overlaps almost perfectly with the Eulerian reference (red solid). The temporal oscillations and noise that were present in the scattered case have been largely smoothed out by the re-interpolation.

(2) The volume-averaged Lagrangian estimate (blue dashed), while improved, still exhibits a systematic offset from the Eulerian reference, particularly in the quasi-steady regime.

4.2.2 RMS error across interpolation combinations

The RMS error heatmap (fig. 3) confirms these observations quantitatively:

(1) **Dramatic improvement in centroid-based accuracy:** After re-interpolation, the centroid-based RMS errors drop to $E_{\mathrm{RMS}} \approx 1.75$–$2.73 \times 10^{-3}$, representing a factor of 3–5 improvement over the scattered case ($E_{\mathrm{RMS}} \approx 8.7 \times 10^{-3}$).

(2) **Modest improvement in volume-averaged accuracy:** The volume-averaged RMS errors decrease only slightly to $E_{\mathrm{RMS}} \approx 2.25$–$2.31 \times 10^{-2}$, compared with $E_{\mathrm{RMS}} \approx 2.93$–$2.98 \times 10^{-2}$ in the scattered case. This limited improvement is consistent with the observation that the velocity field carries the dominant error, and re-interpolation does not fully recover the velocity information lost during particle tracking.

(3) **Interpolation scheme now matters:** Unlike the scattered case, the choice of interpolation scheme has a discernible effect after re-gridding. Cases with cubic interpolation for α_{g} yield lower centroid-based errors (1.75 and 1.90×10^{-3}) compared with linear-α_{g} cases (2.56 and 2.73×10^{-3}), suggesting that higher-order interpolation of the gas fraction field preserves the sharp bubble interface more faithfully during tracking.

4.2.3 Field profile diagnostics

Figure 4 shows the gas fraction and velocity profiles extracted from the re-interpolated fields. Compared with the scattered profiles in fig. 2:

- The gas fraction profiles are now perfectly smooth and overlay the Eulerian reference with minimal deviation. The re-interpolation via natural-neighbour interpolation effectively reconstructs the sharp interface.
- The velocity profiles, while smoother than their scattered counterparts, still exhibit visible deviations from the Eulerian reference, particularly near the bubble boundary. This persistent velocity error explains why the volume-averaged rise velocity remains less accurate even after re-interpolation.

4.3 Summary of findings

Table 2 collects the key RMS error values across all configurations.

Table 2: Summary of RMS errors ($\times 10^{-3}$) for all configurations. Best values in each category are shown in bold.

		Scattered		**Interpolated**	
α_{g}	V_z	Centroid	Vol-avg	Centroid	Vol-avg
Linear	Cubic	8.70	29.4	2.56	22.8
Cubic	Cubic	8.71	29.8	**1.75**	**23.1**
Linear	Linear	8.78	29.3	2.73	22.5
Cubic	Linear	8.79	29.7	1.90	22.8

The following conclusions can be drawn:

(i) **Centroid-based > Volume-averaged:** Regardless of data representation and interpolation scheme, the centroid-based rise velocity is always more accurate than the volume-averaged counterpart. This advantage arises because the centroid depends only on α_{g}, which is interpolated more accurately than V_z.

(ii) **Interpolated > Scattered (for centroid-based):** Re-interpolation onto a fixed grid reduces the centroid-based RMS error by a factor of 3–5. The natural-neighbour interpolation acts as a spatial smoothing filter that recovers the structured representation of the gas fraction field, improving the centroid calculation.

(iii) **Interpolation provides limited benefit for volume-averaged:** The volume-averaged method shows only a 20–25% reduction in RMS error after re-interpolation, because the dominant error source (velocity field) is not substantially improved by the re-gridding process.

(iv) **Velocity is the dominant error source:** Profile comparisons confirm that the gas fraction field is reconstructed with high fidelity from the Lagrangian particles, while the velocity field carries significantly larger errors, both in scattered and re-interpolated forms.

(v) **Cubic α_g interpolation helps after re-gridding:** When re-interpolation is applied, using cubic interpolation for the gas fraction during tracking yields measurably lower centroid-based errors compared with linear α_g interpolation. This effect is masked when using scattered data directly.

(vi) **Recommendation:** For practitioners using Lagrangian particle data to estimate bubble rise velocity, the centroid-based method applied to re-interpolated data with cubic α_g interpolation provides the most accurate results. If re-interpolation is not feasible, the centroid-based method on scattered data is still preferable over the volume-averaged approach.

5 Conclusions

A systematic numerical analysis has been conducted to assess the accuracy of bubble rise velocity estimation from Lagrangian particle tracking data, using a 2D axisymmetric rising bubble at Re = 100, Bo = 100 as the test case. The analysis considered two rise velocity definitions (centroid-based and volume-averaged), two data representations (scattered and re-interpolated), and four interpolation scheme combinations for the gas fraction and velocity fields.

The centroid-based rise velocity consistently outperforms the volume-averaged method, with RMS errors that are 3–10 times smaller depending on the configuration. This superior accuracy is attributed to the centroid method's exclusive dependence on the gas fraction field, which is interpolated more faithfully than the velocity field during particle tracking.

Re-interpolation of scattered Lagrangian data onto a fixed Eulerian grid dramatically improves centroid-based estimates (by a factor of 3–5) but offers limited benefit for the volume-averaged method, whose accuracy is fundamentally constrained by velocity field interpolation errors.

The dominant source of error in the Lagrangian framework is the velocity field interpolation, as confirmed by direct comparison of gas fraction and velocity profiles along diagnostic cut-lines. This observation motivates the use of higher-order interpolation schemes (such as fifth-order Lagrangian polynomials) for the velocity field in future tracking simulations, which is currently being investigated.

$$V_{\text{rise}}^{\text{centroid}} = \frac{d}{dt}\left(\frac{\int_\Omega \alpha(\mathbf{x},t)\, y\, dA}{\int_\Omega \alpha(\mathbf{x},t)\, dA}\right) \tag{10}$$

$$V_{\text{rise}}^{n} = \frac{y_c^n - y_c^{n-1}}{\Delta t}, \qquad y_c^n = \frac{\sum_i \alpha_i^n\, y_i\, \Delta A_i}{\sum_i \alpha_i^n\, \Delta A_i} \tag{11}$$

$$V_{\text{rise}}^{\text{vol-avg}} = \frac{\int_\Omega \alpha(\mathbf{x},t)\, v_y(\mathbf{x},t)\, dA}{\int_\Omega \alpha(\mathbf{x},t)\, dA} \tag{12}$$

$$V_{\text{rise}}^{n} = \frac{\sum_i \alpha_i^n\, (v_y)_i^n\, \Delta A_i}{\sum_i \alpha_i^n\, \Delta A_i} \tag{13}$$

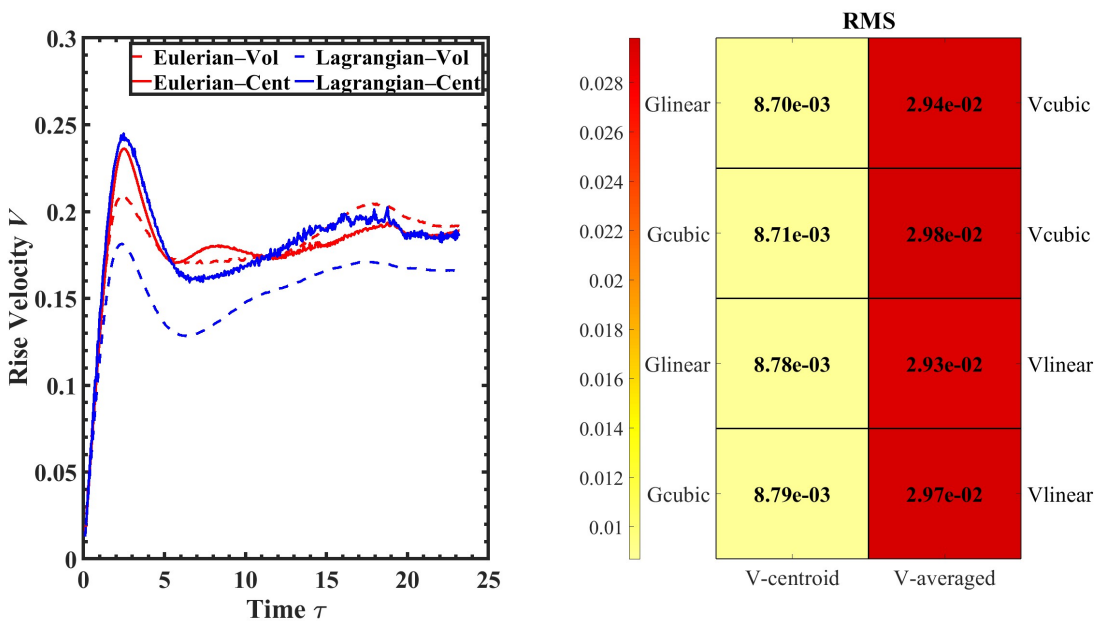

Figure 1: Scattered Lagrangian data (Approach A). **Left:** Temporal evolution of the non-dimensional rise velocity, comparing the Eulerian reference (red) with the scattered Lagrangian estimate (blue) for the first interpolation case. Solid lines correspond to the centroid-based method; dashed lines to the volume-averaged method. **Right:** RMS error heatmap across all four interpolation combinations. Rows indicate the gas fraction interpolation scheme and columns distinguish between centroid-based and volume-averaged methods. Labels on the right margin identify the velocity interpolation scheme.

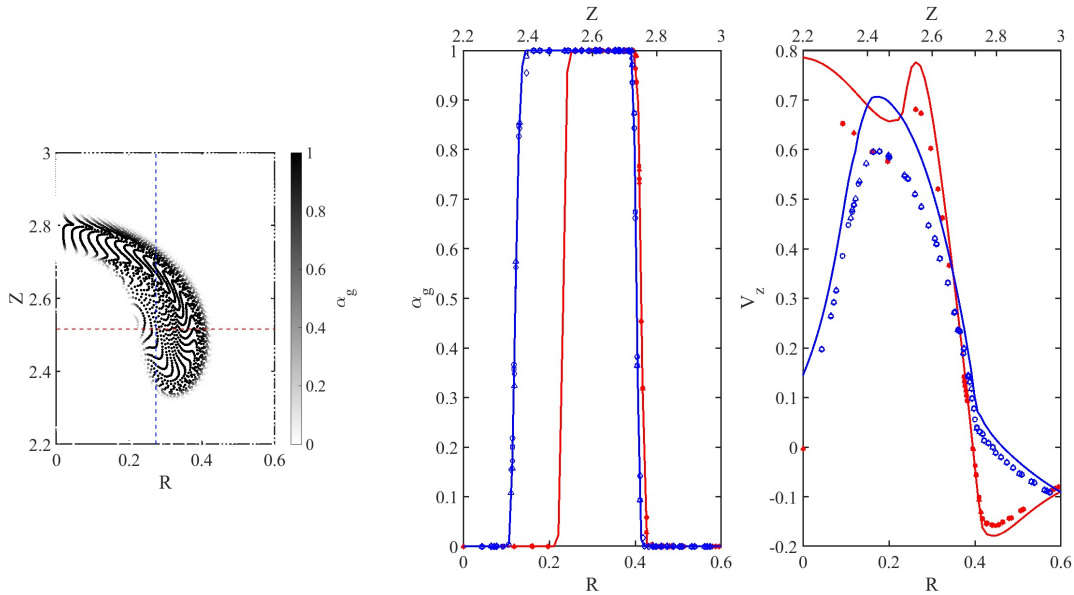

Figure 2: Scattered Lagrangian data (Approach A) at $\tau \approx 6.3$. **Left:** Scatter plot of Lagrangian particle positions coloured by the gas volume fraction α_g; dashed lines indicate the horizontal (red, r-direction) and vertical (blue, z-direction) cut-lines. **Middle:** Gas fraction profiles along the two cut-lines: Eulerian reference (solid lines) versus scattered Lagrangian values (markers) for all four interpolation combinations. Red denotes the r-direction (bottom axis); blue denotes the z-direction (top axis). **Right:** Corresponding axial velocity profiles. The larger scatter in velocity compared with gas fraction confirms that V_z interpolation is the dominant error source.

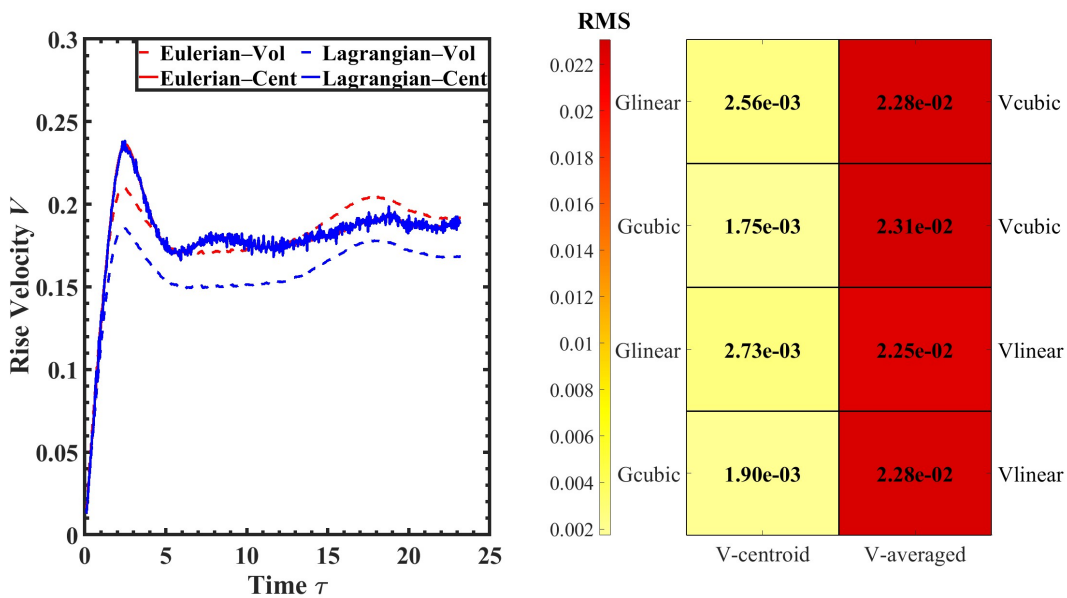

Figure 3: Re-interpolated Lagrangian data (Approach B). **Left:** Temporal evolution of the non-dimensional rise velocity after re-interpolation. The centroid-based Lagrangian estimate (blue solid) now nearly overlaps the Eulerian reference (red solid), while the volume-averaged estimate (blue dashed) remains offset. **Right:** RMS error heatmap. Centroid-based errors have decreased by a factor of 3–5 compared with fig. 1; volume-averaged errors show only modest improvement.

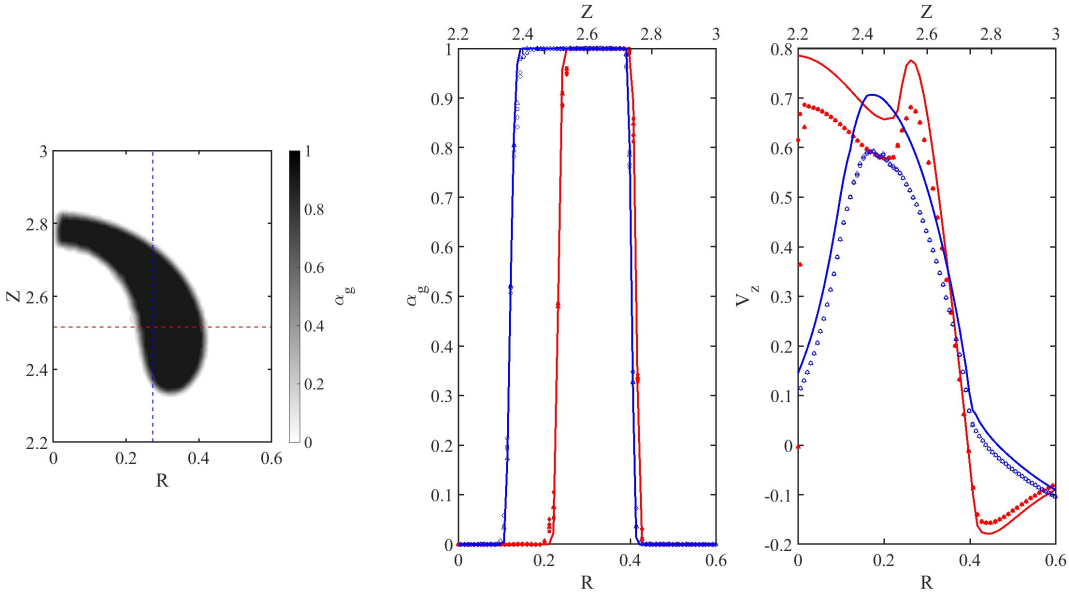

Figure 4: Re-interpolated Lagrangian data (Approach B) at $\tau \approx 6.3$. **Left:** Contour plot of the re-interpolated gas fraction field for the first Lagrangian case, with diagnostic cut-lines indicated. **Middle:** Re-interpolated gas fraction profiles along the cut-lines, now smooth and closely matching the Eulerian reference. **Right:** Re-interpolated velocity profiles. While smoother than the scattered counterpart (fig. 2), deviations from the Eulerian reference persist, particularly near the bubble boundary.